

Power Series and Interval of Convergence



Finding the radius of convergence

- 1 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.
 - 2 Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
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Finding the interval of convergence (testing endpoints)

- 3 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.
 - 4 Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2}$.
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Representing functions as power series

- 5 Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.
 - 6 Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.
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Differentiating and integrating power series

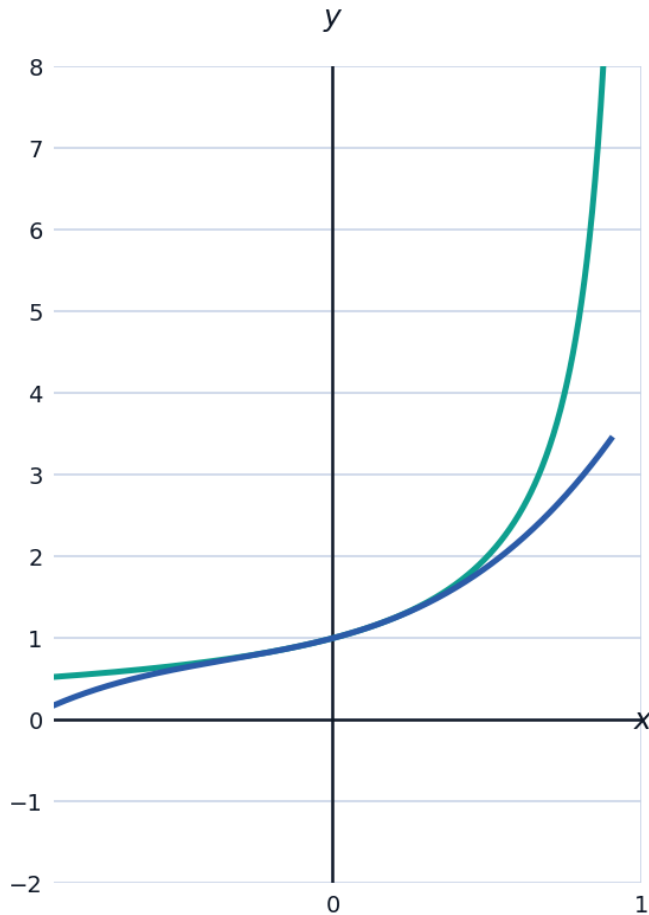
- 7 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.
 - 8 Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.
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More radius/interval of convergence practice

- 9 Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n! x^n$.
 - 10 Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n (x+1)^n}{n \cdot 3^n}$.
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Visualizing a power series approximation

- 11 The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.



Term-by-term operations preserve the radius of convergence

- 12 State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.
- 13 Given that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.

Building a new series by combining known ones

- 14 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.

Ratio Test with factorial and power combinations

15 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n x^n}{n^2}$.

16 For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.

Using a power series to approximate a definite integral

17 Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.